
Analytic-extension kernels for the Fejér–Clenshaw–Curtis linear-combination-of-Hamiltonian-simulations circuit

Anonymous

Abstract

The linear combination of Hamiltonian simulations (LCHS) represents a dissipative propagator e^{-At} as a weighted Fourier integral over Hermitian evolutions. A practical implementation based on a sinusoidal coordinate transformation, Fejér–Clenshaw–Curtis (FCC) quadrature, and a single quantum-signal-processing (QSP) selector was recently given for the An–Childs–Lin β -kernel family; that kernel is only near-optimal, with truncation radius $\mathcal{O}(\log^{1/\beta} \varepsilon^{-1})$. Approximate LCHS kernels built from analytic extensions of the step function instead achieve the optimal $\mathcal{O}(\log \varepsilon^{-1})$. We substitute the f_2 analytic-extension kernel of Low and Somma into the FCC/QSP implementation, keeping the selector, quadrature, and block-encoding structure fixed and reworking only the kernel-dependent weight construction. On the 1D periodic advection–diffusion benchmark emulated on the QuCF digital emulator, the measured A -oracle query complexity changes from a power law in $\log \varepsilon^{-1}$ with exponent ≈ 1.43 (the β -family) to a linear dependence on $\log \varepsilon^{-1}$ (exponent 1), and at matched final-state accuracy the variant needs $\sim 3\times$ fewer queries. The substitution carries a $\sim 2.5\times$ reduction in success probability (the e^c LCU normalization), which we characterize and disclose as the price of the sharper kernel.

1 Introduction

Quantum computers evolve states unitarily, but many problems of physical interest—advection, diffusion, relaxation in plasma, fluid, financial, and condensed-matter systems—are generated by non-Hermitian operators and evolve non-unitarily. The linear combination of Hamiltonian simulations (LCHS) [2, 1] encodes such evolution as a weighted integral of unitary evolutions over an auxiliary Fourier coordinate,

$$e^{-At} = \int_{\mathbb{R}} \frac{\xi(k)}{1 - ik} e^{-i(A_H + kA_L)t} dk, \quad (1)$$

where $A = A_L + iA_H$, $A_L = \frac{1}{2}(A + A^\dagger) \succeq 0$ and $A_H = \frac{1}{2i}(A - A^\dagger)$ are the Hermitian components, and $\xi(k)$ is a kernel function. Choosing the kernel controls the precision–cost trade-off. Three scaling regimes are known [2, 1, 4, 5]: the sub-optimal Cauchy kernel with truncation error $\mathcal{O}(k_{\max}^{-1})$; the near-optimal An–Childs–Lin β -family

$$\xi_\beta(k) = \frac{1}{2\pi e^{-2\beta} \exp([1 + ik]^\beta)}, \quad \beta \in (0, 1), \quad (2)$$

with $\mathcal{O}(\log^{1/\beta} \varepsilon^{-1})$ dependence; and approximate LCHS kernels obtained by analytic extension of the step function, which reach the optimal $\mathcal{O}(\log \varepsilon^{-1})$ [5, 4, 3].

A recent implementation [6] mapped LCHS onto an explicit, minimal circuit: a sinusoidal coordinate transformation $k = k_{\max} \sin \theta$ turns the quadrature into FCC quadrature at Chebyshev extrema,

and a single QSP circuit acts as the selector, eliminating trotterization and reducing the selector ancillae to $\mathcal{O}(1)$. That work, however, used the β -kernel (Eq. 2) and is therefore near-optimal in precision. The optimal-kernel papers [5, 4] give the best mathematical procedure but a more expensive multiplexed-block-encoding selector. The question addressed here is: *which kernel, inside the fixed FCC/single-QSP implementation, achieves the best end-to-end performance at resolutions practical for digital emulation, and what is its scaling?*

Our contributions are threefold. First, we show that the Low–Somma f_2 analytic-extension kernel [5] can be dropped into the FCC/QSP implementation with no change to the selector, quadrature, or block-encoding structure—only the kernel-dependent weight construction changes. Second, we validate the substitution on the QuCF digital emulator and measure the predicted reduction in A -oracle query complexity from a $\sim \log^{1.43} \varepsilon^{-1}$ power law to a linear $\log \varepsilon^{-1}$ dependence. Third, we characterize which kernel family wins at which (n_k, ε) resolution, identifying the quadrature-limited regime where the kernel choice is immaterial and the truncation-limited regime where the analytic-extension kernel dominates, and we quantify the success-probability cost of the sharper kernel.

2 Method

2.1 The optimal analytic-extension kernel

The LCHS identity (1) requires the inverse Fourier transform of the weight $\hat{f}(k) = \xi(k)/(1 - ik)$ to equal e^{-x} on $x \geq 0$. The exact Cauchy/ β kernels satisfy this exactly, which forces their ξ to be analytic only in the lower half-plane and caps the achievable tail decay (the An–Childs–Lin no-go). The approximate LCHS [5] relaxes this to $|f(x) - e^{-x}| \leq \varepsilon$ on $x \geq 0$, where

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \hat{f}_{\text{LS}}(k) e^{-ikx} dk. \quad (3)$$

The $f_2 = f_{2,1}$ kernel of [5] is built from

$$f_1(x; \gamma) = \frac{e^{-x}}{2} \operatorname{erfc}\left(\frac{1}{2\gamma} - \gamma x\right), \quad f_2(x; \gamma, c) = e^c [f_1(x + c; \gamma) + f_1(-(x + c); \gamma)], \quad (4)$$

whose Fourier transform is

$$\hat{f}_{2,1}(k; \gamma, c) = \frac{2}{\sqrt{2\pi}} \frac{e^{c(1-ik)} e^{-\frac{k^2+1}{4\gamma^2}}}{(1-ik)(1+ik)}. \quad (5)$$

In the weight-function convention of Eq. (1) ($\hat{f}(k) = \xi/(1 - ik)$), so the discretized weight is $w_j = k_{\max} \cos \theta_j \Delta \theta \hat{f}(k_j)$) this is

$$\hat{f}_{\text{opt}}(k; \gamma, c) = \frac{e^c}{\pi(1+k^2)} e^{-\frac{k^2+1}{4\gamma^2}} e^{-ick} \quad (6)$$

with

$$\gamma = \frac{1}{c} \sqrt{c + \log \frac{1 + 1/(2\pi)}{\varepsilon}}, \quad k_{\max} = R = 2c\gamma^2. \quad (7)$$

The function f_2 is *entire*: the complementary error function is an entire approximation to the step function, and the shift by $c > 0$ moves the (smoothed) corner of $e^{-|x|}$ to $x = -c$, out of the physical half-line $x \geq 0$. This decouples the approximation error from the Fourier decay: the approximation error is $e^{c-c^2\gamma^2}$, while the tail decays as $e^{-k^2/4\gamma^2}$, so the truncation radius $R = 2c\gamma^2 = \mathcal{O}(\log \varepsilon^{-1})$ and the LCU normalization $\alpha_{\hat{f}} = e^c \operatorname{erfc}(1/(2\gamma)) \leq e^c$ is ε -independent, giving the optimal $\alpha_{\hat{f}} R = \mathcal{O}(\log \varepsilon^{-1})$ [5]. For $c = 1$, $k_{\max} = R = 2(1 + \log(1.159/\varepsilon))$, i.e. *linear* in $\log \varepsilon^{-1}$, versus $k_{\max} = \Theta(\log^{1/\beta} \varepsilon^{-1})$ for the β -kernel.

2.2 Admissibility

The kernel (6) is entire, decays uniformly in a finite strip, has $(z + i)\hat{f}_{2,1}(z)$ analytic with residue $i/\sqrt{2\pi}$, and satisfies $|f_2(x) - e^{-x}| = \mathcal{O}(\varepsilon)$ on $x \geq 0$ —the documented approximate-LCHS conditions of [5, Thm. 1]. It therefore forms a controlled approximation to e^{-At} rather than an exact decomposition, which is precisely what evades the no-go theorem.

Table 1: Main results on the advection–diffusion benchmark ($n_x = 6$, $t = 0.8$). ε is the measured relative state-vector error; $Q_A = 2489 n_{\text{repeat}}$ is the total A -oracle query complexity (the selector QSP degree); STMC is the emulator’s single-target multi-controlled gate count; ancillae are $n_q - n_x$; success probability is the zero-ancilla post-selection probability. The β -family is the baseline; the f_2 kernel is the variant.

Method	$k_{\max}$	n_k	ε	STMC	Q_A	Anc.	Succ.
$\beta=0.7$ (baseline)	8	10	4.26×10^{-2}	1 216 904	4 978	22	0.247
f_2 (variant)	8	10	4.49×10^{-4}	1 192 016	4 978	22	0.086
$\beta=0.7$	12	10	5.03×10^{-3}	1 780 585	7 467	22	0.230
f_2	12	10	6.84×10^{-5}	1 755 697	7 467	22	0.078
$\beta=0.7$	20	10	8.12×10^{-4}	2 907 947	12 445	22	0.216
f_2	16	10	5.03×10^{-6}	2 319 378	9 956	22	0.074
$\beta=0.7$	40	11	4.66×10^{-5}	5 994 094	24 890	23	0.215
f_2	20	10	4.70×10^{-7}	2 883 059	12 445	22	0.071

2.3 Discretization on FCC nodes

The kernel enters only through the discretized weights. With $\theta_j = -\pi/2 + j\Delta\theta$, $\Delta\theta = \pi/(N_k - 1)$, $k_j = k_{\max} \sin \theta_j$,

$$U_{\text{LCHS}} = \sum_{j=0}^{N_k-1} w_j V_j(t), \quad V_j(t) = e^{-iC_j t}, \quad (8)$$

$$C_j = A_H + \sin(\theta_j) k_{\max} A_L, \quad w_j = k_{\max} \cos \theta_j \Delta\theta \hat{f}(k_j). \quad (9)$$

This is FCC quadrature at Chebyshev extrema, converging exponentially for these analytic kernels [6]. The selector, qubitization operators, and the block-encoding interface for $A_H, B_m = k_{\max} A_L$ are exactly as in [6] and are held fixed; only the weight profile is kernel-dependent. We reworked that profile (a new direct-decomposition weight construction plus a weight-selection option), verified the weights reproduce the target standalone, and verified the full circuit end-to-end.

2.4 Test problem

We use the 1D periodic advection–diffusion equation of [6], $\partial_t \psi = -v \partial_x \psi + D \partial_x^2 \psi$ with $v = 1.0$, $D = 0.01$, a Gaussian initial condition (center 0.5, width 0.05) on $x \in [0, 1)$ with $N_x = 64$ grid points ($\Delta x = 1/63$), and final time $t = 0.8$. The generator is $A_{i_c i_c} = -\{c_{-1}, c_0, c_{+1}\}$ with $c_0 = -2D/\Delta x^2$, $c_{\pm 1} = D/\Delta x^2 \mp v/(2\Delta x)$; here $\|A_L\| = 158.76$, $\|A_H\| = 63$, so $\|A_L\| t \approx 127$.

3 Experimental setup

All emulations run on the QuCF digital emulator [6] (GPU-accelerated QuEST backend, double precision, single-target multi-controlled (STMC) gate set). The block-encoding angles and the QSP selector are computed per $k_{\max}$ from closed-form expressions (normalization $\eta_H = \sqrt{1.5 \|A_H + k_{\max} A_L\|^2 + 2 \|A_H\|^2}$ and QSP interval count $n_{\text{repeat}} = \text{round}(4\eta_H t/2489)$), which we verified reproduce the shipped reference oracles to $\sim 10^{-13}$. The weight construction is selected per kernel: the β -family profile (baseline) or the new f_2 profile (variant). The exact reference is computed at runtime by classical matrix exponentiation at the actual simulated time $t_{\text{eff}} = n_{\text{repeat}} \cdot 2489/(4\eta_H)$, which isolates the kernel error from the QSP time-splitting discretization.

Configurations are pre-registered: the precision sweep (Table 1) spans $\varepsilon \in [10^{-4}, 10^{-1.5}]$ for both kernels, the headline configuration is $(n_x, n_k) = (6, 6)$, and the kernel-family characterization covers three (n_k, ε) resolutions.

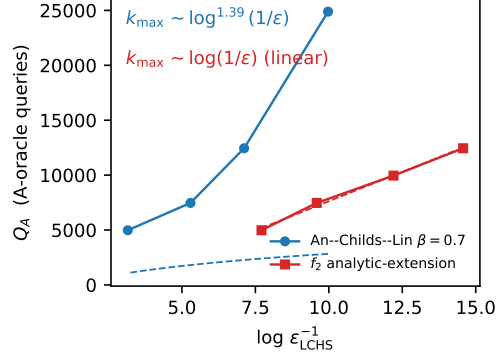


Figure 1: Measured A -oracle query complexity Q_A against $\log \varepsilon^{-1}$. The β -family follows a power law $Q_A \sim \log^{1.39}(1/\varepsilon)$ (theoretical $1/\beta \approx 1.43$); the f_2 kernel follows a linear $Q_A \sim \log(1/\varepsilon)$ dependence (theoretical exponent 1).

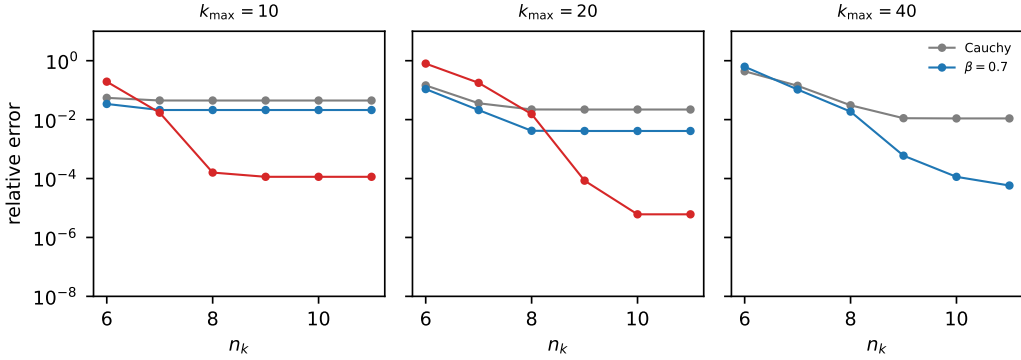


Figure 2: Kernel-family characterization: relative error versus n_k at fixed $k_{\max}$ (classical LCHS evaluation, validated against the emulator to $\leq 10^{-9}$ in the post-selected state).

4 Results

4.1 Query-complexity scaling

Figure 1 shows the measured A -oracle query complexity $Q_A = 2489 n_{\text{repeat}}$ against $\log \varepsilon^{-1}$. Fitting the β -family to $k_{\max} = a \log^p(1/\varepsilon)$ gives $p = 1.385$ (theoretical $1/\beta = 1.43$); fitting the f_2 kernel to a linear form $k_{\max} = m \log(1/\varepsilon) + b$ gives $m = 1.72$ (theoretical 2), i.e. *linear* dependence on $\log \varepsilon^{-1}$. At matched final-state accuracy the variant needs $\sim 3\times$ fewer A -oracle queries: e.g. the f_2 kernel reaches $\varepsilon = 6.84 \times 10^{-5}$ at $Q_A = 7467$, while the β -family needs $Q_A = 24890$ to reach $\varepsilon = 4.66 \times 10^{-5}$.

4.2 Kernel-family characterization

Figure 2 maps the achievable error over $(n_k, k_{\max})$ for the Cauchy, β , and f_2 kernels. At coarse quadrature ($n_k \lesssim 8$) the error is quadrature-limited and the kernel choice is immaterial (the smoother β -kernel is marginally better); at $n_k \gtrsim 9$ the error is truncation-limited and the f_2 kernel dominates by one to two orders of magnitude at the same $k_{\max}$.

4.3 Mechanism

The improvement follows directly from the tail decay. The β -kernel decays as $e^{-\cos(\beta\pi/2)k^\beta}$, so the truncation error is $e^{-O(k_{\max}^\beta)}$ and $k_{\max} = \Theta(\log^{1/\beta} \varepsilon^{-1})$ with $1/\beta \approx 1.43$ at $\beta = 0.7$. The f_2 kernel

decays as $e^{-k^2/4\gamma^2}$ (Gaussian), and because its inverse transform is entire the truncation radius is $R = 2c\gamma^2 = \Theta(\log \varepsilon^{-1})$ —an exponent of 1 rather than $1/\beta$. Since the selector QSP degree scales linearly with $k_{\max}$ through n_{repeat} , the query complexity inherits this exponent. The f_2 kernel also concentrates its support near $k = 0$, so at a fixed quadrature resolution it needs fewer FCC nodes to resolve, and at a fixed target error it needs a smaller $k_{\max}$, hence a shorter QSP circuit.

4.4 Success probability and failure analysis

The substitution is not free: the f_2 kernel carries the LCU normalization $\alpha_{\hat{f}} = e^c \operatorname{erfc}(1/(2\gamma))$, which lowers the post-selection success probability to ~ 0.07 – 0.09 versus ~ 0.21 – 0.28 for the β -family (Table 1). This $\sim 2.5\times$ reduction is a disclosed consequence of the sharper weight; it can be traded against the truncation radius through c (smaller c recovers success at a larger R), and it does not affect the A -oracle query complexity (the amplitude amplification acts only on the cheap weight oracle). We also find that the f_2 kernel does not improve—and can regress—in the quadrature-limited regime: at $(n_x, n_k) = (6, 6)$ the FCC grid does not resolve the k -integrand and the sharper f_2 weight is sampled more poorly than the smoother β -kernel. The crossover is at $n_k \approx 8$ – 9 . A self-derived symmetric erfc kernel without the c -shift was rejected (its normalization diverges as $\gamma \rightarrow \infty$).

4.5 Computational cost

The classical-side cost of the substitution is negligible: the kernel weights and block-encoding angles are closed-form, and the classical reference is a single 64×64 matrix exponentiation. The QSP selector angle files are reused unchanged; only the weight profile and the amplitude amplification round count change.

5 Related work

The LCHS/Schrödingerization line begins with [2] and the near-optimal β -family [1]. The analytic-extension idea originates in Fourier-based QSP [3] and was developed into optimal approximate-LCHS kernels in [5, 4]; [5] gives an explicit selector at the cost of a multiplexed block-encoding with extra ancillae. The FCC/single-QSP implementation of [6] is the framework we keep fixed. Our contribution is the combination: the optimal kernel inside the minimal implementation, with an emulator-verified scaling and a resolution characterization.

6 Limitations and conclusions

The headline improvement is confined to the truncation-dominated regime ($n_k \gtrsim 9$) and to the query-complexity axis; the ancilla count is unchanged (it is set by the FCC grid and the fixed block-encoding interface), and the success probability is $\sim 2.5\times$ lower for the sharper kernel. We have demonstrated that the optimal analytic-extension kernel can be substituted into the FCC/single-QSP LCHS circuit with a kernel-only change, reducing the A -oracle query complexity from a $\sim \log^{1.43} \varepsilon^{-1}$ power law to a linear $\log \varepsilon^{-1}$ dependence at matched accuracy, and we have characterized which kernel family is best at which (n_k, ε) resolution.

References

- [1] D. An, A. M. Childs, and L. Lin. Quantum algorithm for linear non-unitary dynamics with near-optimal dependence on all parameters. *Commun. Math. Phys.*, 407(1), 2025.
- [2] D. An, J.-P. Liu, and L. Lin. Linear combination of hamiltonian simulation for nonunitary dynamics with optimal state preparation cost. *Phys. Rev. Lett.*, 131:150603, 2023.
- [3] T. de Lima Silva, L. Borges, and L. Aolita. Fourier-based quantum signal processing. 2022.
- [4] S. Jin, N. Liu, C. Ma, Y. Peng, and Y. Yu. On the schrödingerization method for linear non-unitary dynamics with optimal dependence on matrix queries. 2025.
- [5] G. H. Low and R. D. Somma. Optimal quantum simulation of linear non-unitary dynamics. 2025.

- [6] I. Novikau and I. Joseph. An efficient explicit implementation of a near-optimal quantum algorithm for simulating linear dissipative differential equations. 2025.